An Evolutionary Classification of the Strategies for Drug Discovery

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ABSTRACT

The aim of this paper is to provide a framework for understanding how strategies for drug discovery are selected and sustained by their environment. This framework is based on a classification of strategies with reference to their evolutionary attributes. During the last century, pharmaceutical research has evolved from a pattern where random discoveries dominated research to a more rational and planned one, where the outcome of the research is almost statistically determined. This change has resulted in the reduction of the amount of risk and serendipity involved in the process. The paper will provide a historical synopsis to examine how this evolution occurred. It will then provide a classification of the strategies for drug discovery using the biological classification technique *cladistics*. The attributes of a fit strategy within the pharmaceutical environment will be discussed and it will be argued that almost all the emergent strategies of the past (high serendipity) are rapidly being replaced by less serendipitous planned ones. Finally, the paper will conclude by discussing the key elements of a successful strategy for drug discovery.

INTRODUCTION

The need to provide adequate medicines has been present for as long as humanity can trace its roots. Since antiquity people have tried to cure diseases by eating, drinking or applying substances; often plant extracts that are now referred to as herbal medicines. Originally, the attempts to discover new drugs and new cures were based on intuition and empirical observation, while any successful results were more likely to be the product of fortunate accidents. Over the last century however, the search for new drugs has changed dramatically. Advances in the understanding of human biology, new technology, and more recently market considerations have altered the way drugs are being discovered. Pharmaceutical organisations have developed into large corporations, which are not prepared to rely upon 'fortunate accidents', nor can they afford to. This has resulted in pharmaceutical organisations seeking different strategies to optimise the efficiency of their drug discovery process.

This endeavour to optimise the drug discovery process has resulted in the progressive replacement of the highly serendipitous process of drug discovery with a process dominated by new technology where scientists can almost statistically predict the outcome of their research. This change was primarily the result of the strategic imperative and the significant technology changes that occurred during the last fifty years (Ratti and Trist, 2001)

Therefore, one of the key questions to be addressed is: What is the mechanism that facilitates the progress of strategies for drug discovery? This paper seeks to answer this by using the concept of evolution. Evolutionary literature suggests that for entities such as biological species to follow an evolutionary pattern they must satisfy four criteria: *variation, heredity*, *selection* and existence in *populations*. This paper will demonstrate that strategies for drug discovery also conform to these criteria and therefore it may be argued that they follow an evolutionary pattern.

In summary, the aim of this paper is to explore and present the evolutionary characteristics of drug discovery strategies. To achieve this it will begin by presenting a brief history of the drug discovery process. This starts with the *age of botanicals* (pre 1800s) and ends with today's highly sophisticated and highly automated era. This will be followed by a definition and discussion of the term strategy and how it relates to drug discovery within the context of this paper. The paper will then discuss evolution and the requirements that most entities must satisfy to exist and change as an evolutionary entity. The paper will then explain how strategies for drug discovery meet the requirements of evolution by using an evolutionary classification to show different strategies and their defining characteristics. The paper will conclude by arguing that biological evolution can explain the change of the strategies for drug discovery.

HISTORY OF DRUG DISCOVERY

As mentioned in the introduction of this paper, the process of discovering new drugs has changed and with dramatic changes occurring in the last century. This section will present a summary of the history of drug discovery to aid the reader in understanding the evolutionary arguments presented in this paper.

Pre 1800 to 1919

The pharmaceutical years up to the 1800s are called *the age of botanicals* (Somberg, 1996). Since the start of human history, there are reports of man making use of herbals for foods, items of religious significance and as medication. In Greece and Egypt evidence of pharmacopoeia has been found which include compounding and dosage requirements for various illnesses (Bogner, 1996). Later, during the first half of the 19th century, diseases were identified by symptom and the road to health was to attack the symptom as vigorously as possible (Lesney, 2000). During the second half of this century, the patent medicines and in particular homeopathy became the basis for a commodity based medicine industry that in turn gave birth to the pharmaceutical industry. Among the drugs that were developed, a few authentic drugs were accidentally discovered: quinine, digitalis, and cocaine and close to the end of the century antipyrine and aspirin.

This era provided modern drug discovery with knowledge that led to the understanding of the biological activity of some substances found in nature. In addition, the acceptance of the 'microbial theory of disease' helped to explain how infectious diseases are caused by microscopic living agents. This theory was a key development in medical science.

1920s and 1930s

The main characteristics of these two decades were the discovery of vitamins and developments in chemistry (Pizzi, 2000). Also, new vaccines and new drugs were developed, with one of the most important, being the almost accidental discovery of penicillin in 1928 by Alexander Fleming. Finally, another contribution of this era was the interdisciplinary thinking that led to the development of new medical instruments and associated technology. For the first time knowledge acquired in biology, physics and chemistry was combined to advance the field of pharmaceutical science.

1940s

This era is commonly known as the *antibiotic era*. During these years, drugs were discovered in a less serendipitous fashion. Researchers started looking for specific drugs and often managed to find them. This resulted in the discovery of numerous drugs that were targeted at various diseases. Also, World War II played a very important role. For example, the need to cure infected soldiers accelerated the development and production of penicillin and antibiotics.

1950s

Drug discovery during this decade was also influenced by world events such as the Cold War, the Korean conflict, the launch of the first orbital satellite in 1957, etc. Technologies previously used for other scientific purposes or for warfare were now being used for civilian needs. New technology and new instrumentation coupled with an understanding of how the human body worked and the understanding of the DNA's structure, opened new windows of opportunity for the development of new drugs. Breakthroughs were made in the instrumentation that led to the development of the biotechnology. The main characteristics and contributions of this decade were mainly the vast amount of knowledge about human biology and chemistry, the development of sophisticated instrumentation, and the shift of the drug discovery process to a less serendipitous pattern.

1960s

The 1960s was the pharmaceutical decade of the century (Frey and Lesney, 2000). During this period, people became conscious about *pills* in all aspects of their lives. Analytical chemists and biologists worked together, as never before, in the search for new drugs. Knowledge and insights about DNA and the acceptance that it was indeed genetic material would mark the years to follow. Also, the enhancement of instrumentation and the introduction of powerful computing technology facilitated laboratory automation and aided the discovery and development of new drugs. Finally, breakthroughs in etiology (study of the causes of the disease) increased understanding and helped in the development of science driven strategies for the discovery of new drugs. This caused a significant shift away from the serendipitous approach.

1970s

During this decade, the War against cancer began. Computing technology was coupled with the life sciences, and genetic engineering emerged. In 1976, Genentech Inc., in San Francisco USA, was founded in an attempt to capitalise on new discoveries and this marked the birth of the biotechnology industry. The 70s also saw the dawning of rapid developments in computing power and applications. Finally, during the 1970s the knowledge and instrumentation that was in place that eventually led to rational drug design in the 1980s and 1990s

1980s

This decade was characterised by further scientific and technological developments and the appearance of new diseases, with AIDS (Acquired Immune Deficiency Syndrome) being the most significant. AIDS motivated the development of immunology i.e. the study of the body's resistance to infections. Another phenomenon that triggered novel research was the resistance of old diseases to conventional drugs.

In the area of drug discovery, the developments that marked this decade revolved around the application of computing technology to the design of new drugs and the strategic acquisition of small biotechnology firms by larger pharmaceutical firms. Molecular biology and the use of computers gave rise to a new approach to innovation. Up to this point in pharmaceutical history, the discovery of active compounds depended on a 'try and see' empirical approach. During this period, knowledge on body physiology and on medical disorder coupled with knowledge on drugs would result in the conceptualisation of the correct molecules. This ideal molecule design is then presented to research chemists who search for a close match.

This decade also saw the commercialisation of drug discovery. Small entrepreneurial biotechnology firms were founded to discover active compounds. However, many of these firms were struggling for survival by the mid 1980s. This led to their acquisition by large pharmaceutical firms. This acquisition of smaller biotechnology firms provided stable source of funding for the former and expertise in a potentially successful area for the latter.

Finally, another very significant step was the development of *combinatorial chemistry* i.e. the simultaneous use of large sets of chemically similar agents in order to produce thousands of organic compounds that are then screened for biological activity.

1990s

Following the advances in the computational technology in the 1970s and 1980s and their use in the health sciences, the 90s saw the use of robotics and automation in drug discovery. Apart from the rapid advances in the technology in drug discovery primarily due to the use of computers this decade was also marked by the continuation of the reappearance of diseases previously treated. During this decade the management aspect of drug discovery also developed in the sense that large amounts of information and scientists from various disciplines had to be effectively managed in order to increase the efficiency of the drug discovery process and reduce its lead time.

The main contributions and characteristics of the final decade of the century are associated with the rationalisation of the drug discovery process. The competitive pressures and the direct link of the number of discoveries with the financial performance of the organisation, further enforced the development of effective drug discovery. In addition, new technologies helped the development of libraries of compounds that would aid the scientists in the drug design.

The 20th century has seen tremendous advances in the medical sciences resulting in the presence of a multibillion-dollar industry. Particularly during the last two decades the combination of the advances in computers and the knowledge in human biology has created a plethora of knowledge and resources. Therefore, the effective utilisation of these resources became a major issue for the pharmaceutical companies to address.

STRATEGIES FOR DRUG DISCOVERY

The history of drug discovery described in the previous section suggests an evolutionary pattern of drug discovery and the strategies associated with it. Both the technology and the management of research have changed remarkably over the previous century. As the understanding of human biology progressed, the research moved from a random pattern to a more predictable and less risky one, where an organisation could almost statistically predict the outcome of the research and its market potentials. This change has been affecting the way organisational strategies are being formed and implemented throughout

the history of the pharmaceutical industry. For instance, drug discovery strategies have been formed by four factors*: scientific knowledge, available technology, human resources* and *diseases*. The scientific knowledge, the technology, and human resources provide the *strategic assets* by which the strategies were formed. The diseases coupled with market factors such as legislation provide the environment where the fittest strategies would succeed.
During the early periods, when knowledge and technology were limited, there was minimal or no strategic planning and the development of new drugs was dominated by scientific fate and chance. As industry matured and knowledge enhanced understanding of both human biology and chemistry, scientists developed a better comprehension of diseases that allowed targeted solutions (*rational drug design*). Also, the creation of large pharmaceutical corporations and the linking of discoveries with organisational profits increased the need for a more controlled and business like approach to discovery.
Figure 1 represents the evolution of strategy from a primarily emergent process to a primarily planned process. In the early 1900s there was limited understanding of the human biology and chemistry and thus the strategies were not focused. As knowledge and understanding increased the strategies became more focused on cure of specific disease using technology and scientific knowledge.

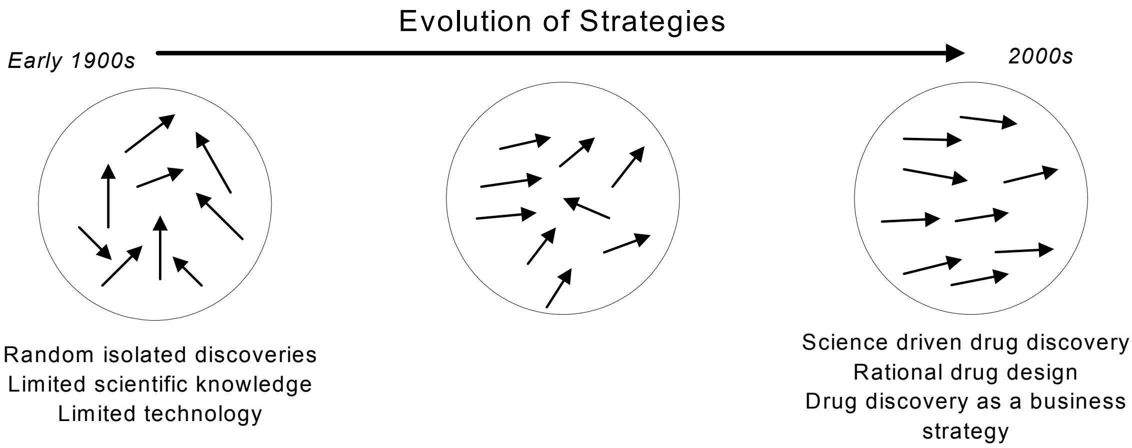

Figure 1, Evolution of strategies for drug discovery

This view is supported by Henderson (1994) who views the drug discovery process as no longer a *'...random procedure in which inspired scientists, working around the clock, come upon breakthroughs in the middle of the night'*. Rather the scientists now have at their disposal much more information and more sophisticated tools that help guide them through the process. Today's drug discovery requires the integration of knowledge from a variety of disciplines including molecular biology, physiology, biochemistry, molecular kinetics, etc.

With this strategic and evolutionary context, this paper proposes the following definition for drug discovery:

The pattern defined or adapted by the management of a pharmaceutical organisation, within a certain environment, to effectively materialise the corporation's goals and policies in achieve a competitive advantage through the application of scientific knowledge, technology and available resources on the discovery of new drugs.

The main points of this definition are:

- Strategy is seen as a pattern, rather than a predetermined plan. Viewing strategy as a plan is not appropriate for drug discovery, as it does not allow for serendipity, a crucial factor in any research activity and particularly in the discovery of new drugs. In contrast, a pattern suggests a more adaptable or evolutionary behaviour and therefore complies with the history of drug discovery that has been highly influenced by external factors such as diseases, technology, and political and social circumstances.
- The strategy for drug discovery is seen as a business or operational strategy that helps to realise the corporate strategy (…materialise corporation's goals…). As discussed earlier a business strategy is concerned with establishing a competitive advantage. The development of a new drug for a specific disease provides an organisation with such an advantage and therefore it would be rational to categorise drug discovery strategies as a business level strategic activity.
- The formation of a strategy for drug discovery is achieved using three strategic assets: *knowledge*, *technology* and *available resources*. The environment within which a drug discovery strategy is formed and executed, is defined by corporate goals and policies i.e. the corporate strategy. Knowledge, technology and available resources are used within the organisation to create and protect the firm's competitive advantage and to meet the goals set by the corporate strategy.

The following sections of this article explore and discuss how strategies for drug discoveries have evolved during the last century by identifying key strategies along with their characteristics.

EVOLUTION OF STRATEGIES FOR DRUG DISCOVERY

To study the evolution of drug discovery strategies it is necessary to examine how evolution occurs. To achieve this, the paper will introduce the processes of evolution, often referring to examples that occur in biological evolution. It is important to note, that even though evolution as a discipline is primarily the domain of biology, it is a process that exists in technical, social and economic systems.

When studying biological organisms evolutionary changes are easier to see. Aquatic worms have evolved into mammals, birds, reptiles, amphibians, and fish over 600,000,000 years and bacteria evolved into plants and animals over 3,000,000,000 years. However, in the case of strategies for drug discovery, evolution is not as evident and knowledge about the pattern of change is extremely limited. To address this issue this paper examines the requirements of evolution as originally developed by the biological sciences, but also recently adapted by organisational scientists such as Hannan and Freeman (1977), and Aldrich (1979). It will demonstrate how drug discovery strategies conform to these requirements.

Requirements for Evolution

Evolution as a term was first introduced at the beginning of the 19th century as a scientific explanation for the existence of the organisms. It was resisted at the time, particularly by theologians who believed that everything was designed by the *Creator*. Now, the evolution of species is more commonly accepted and it underpins most scientific research that seeks knowledge on the different sources of evolution.

One of the first requirements for evolution is that species (or the entity understudy) must exist in *populations* rather than in individual forms. Changes in biological evolution do not happen to individual organisms, but to populations over long periods of time. The frequency of this change depends on the members of the population i.e. the species as well as the environment they inhabit.

Also, for evolution to occur there must be some form of diversity between the members of the population (Strean and Hoekstra, 2000). This is the second requirement for evolution and is referred to as v*ariation*. Within a population of species there are innumerable variations (Hutchison, 1974, p. 78). There are two key causes for this phenomenon within the population, namely *environmental* and *genetic*. According to Hutchinson (1974), the environmental factors are any factors found outside the organism, while genetic are those factors found within the organisms. Both these factors must be present for evolution to occur. If the genetic factors are not present, there would be no changes other than those changes caused by environmental alterations (Ridley, 1996). Variation due to genetic effects causes some organisms to be stronger than others and with limited resources. Thus, it is only successful competitors that will reproduce themselves (Ridley, 1996).

Variation within organisms must happen in two different dimensions for evolution to occur (Ridley, 1996, pp.72). These are i) variations within the members, i.e. different physical characteristics, and ii) variations within the fitness of organisms. The first dimension requires the characteristics of the members of the population to have different physical attributes. For instance if evolution of the body sizes of the population is examined then the members must have different body sizes. The second dimension requires the individuals within the population to have different fitness levels. An individual with higher fitness within a population is one that is more likely to reproduce than others. Therefore the second dimension requires some individuals within a population with some characters to be more likely to reproduce.

Heredity is the third requirement of evolution. It explains the adaptation of a population to its environment. This requirement holds that there must be a way for the physical and mental characteristics to be inherited from one generation to the next. The degree of the population's adaptation depends on how far the differences between individuals that are responsible for their success, are inherited by the next generation (Smith, 1993).

Natural Selection was originally introduced by the English naturalist Charles Darwin in the 19th Century. Given the fact that the environment provides only limited resources, living organisms have to compete for them. Natural selection means that those individuals that are not suited to survive in certain environments do not survive (Hutchison, 1974, p127).

These four requirements, population, variation, heredity, and natural selection are the conditions that are required to deduce that evolution occurs. In simple terms for evolution to take place there must exist a population of various species (not identical to each other but similar) that adapts to its environment over generations by the means of natural selection.

Drug Discovery

The aim of this section is to validate the hypothesis that drug discovery strategies are evolving entities, by providing a framework that will encompass the above issues. This framework will group organisational activities over a period and will allow the study of their change. This framework is an evolutionary classification of the strategies for drug discovery – *cladistics*.

This method of classifying and the description below is based on the work of McCarthy (1995) and McCarthy et al (1997) that first explored the concept of using biological classification techniques for classifying manufacturing organisations. This was then followed by McCarthy and Ridgway (2000) who presented a seven-stage method for constructing a cladistic classification of manufacturing organisations.

The aim of this classification will be as follows:

- To demonstrate variation by grouping the different strategies along with their characteristics
- To reveal the populations of these strategies by observing which strategies are located in the same categories, and
- To explore how the requirement of natural selection is satisfied, by identifying which strategies have been selected in.

The main benefit of using cladistics for classifying drug discovery strategies is that the resulting classification, depicted as a treelike structure called a *cladogram*, would demonstrate that the pattern of evolution and the resulting variation.

RESEARCH METHODOLOGY

The methodology employed by this research follows the steps shown below:

- Construction of conceptual classification
- Collection of empirical data[1]
- Validation of conceptual classification
- Validation of hypothesis

More detailed description on the construction of the cladogram may be found at the following papers McCarthy (1995) and McCarthy et al. (1997), McCarthy and Ridgway (2000), Fernandez et al. (2001), Leseure (2000) McCarthy et al. (2000).

Construction of conceptual classification

The construction of the conceptual classification involves the collection and review of relevant literature and the identification of different drug discovery strategies and their defining characteristics. The various identified strategies and the defining characteristics are grouped in a matrix. Table 1 depicts part of the matrix (the full matrix consists of fourteen pages of data and is available from the authors) that has been developed for this research. Once this matrix is complete,

[1] The status of the research at this point in time has completed the collection of the empirical data but is not yet in a format to be included in this paper

the conceptual cladogram is constructed using appropriate cladistic software that allows the construction of a cladogram. The software used in this case is called PHYLIP). The resulting cladogram is shown in Figure 2.

		Strategies				
		Partnerships/ Acquisitions	Outsourcing	Me-too	R&D concentration for innovation and competition	R&D concentration for niche markets
Characteristics	Robotics/ Automation	0	0	0	0	0
	Proteomics	0	0	0	0	0
	Bioinformatics	0	0	0	0	0
	Mimic old drugs	0	0	1	0	0
	Multinational Companies	1	0	0	1	0
	Medium sized companies	0	0	0	0	1
	Be the first with generics	0	0	0	0	1
	Division of Labour	1	0	0	1	0

Table 1, Section of the strategy/characteristics matrix

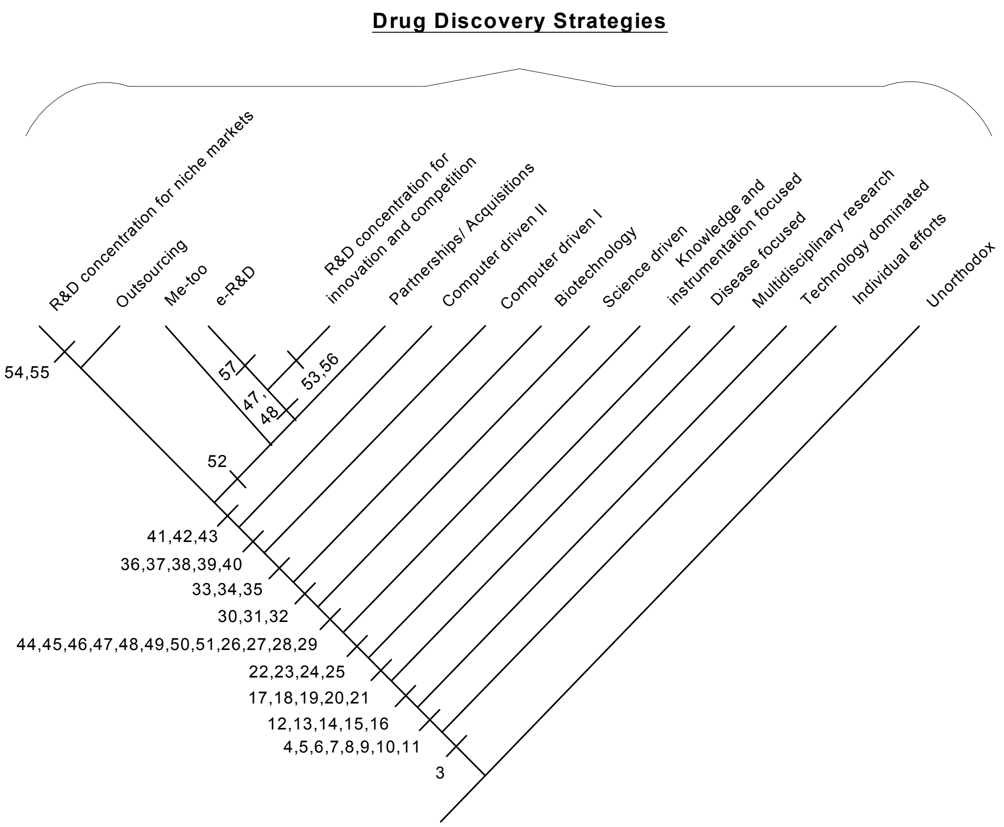

Figure 2. Conceptual Cladogram Showing the Strategies for Drug Discovery

On the right hand side of the cladogram, the more primitive strategies are shown while at the left-hand side the more recent strategies are grouped. The differences between the strategies are the characteristics shown on the branches (lines) of the cladogram. For example, the difference between the *Unorthodox* strategies and the *Individeff* (individual efforts) is *characteristic 1* (*Herbals' use*) i.e. individuals started moving away from unorthodox ways to drug discovery like religious remedies, when the discovered the possible therapeutic use of the herbals.

Collection of empirical data

The data that was collected for the purposes of this research was secondary. It was available from public sources and considered adequate for validation of the theory under question. The pharmaceutical sector is one that has been studied extensively. Therefore, various sources of information exist to provide raw data for analyses and evaluation of pharmaceutical organisations. The sources used for this research include, but are not limited to annual reports, development pipelines (list of products which are under development by individual pharmaceutical organisations), pharmaceutical databases, and historical industry data.

DISCUSSION

This section evaluates Figure 2 to illustrate that drug discovery is an evolving entity.

Existence in populations

The first requirement of evolution requires drug discovery strategies to form populations. Figure 2 shows the classification of similar organisational strategies i.e. drug discovery strategies. Therefore, the overall classification represents one population of strategies. The nature of the strategies within the population varies according to the degree of emergence that the strategies embody. Mintzberg and Waters (2000) also support this argument. They presented a classification of organisational strategies along a *emergent-deliberate* continuum. Although their classification accounts for emergence, its application on drug discovery strategy is not straightforward. The reason for this is that their work focuses more on corporate strategy rather than on business strategy.

Variation

The second requirement for evolution was that of *variation*. Variation is caused by both environmental and genetic conditions. Moreover, variation between members of a population must occur as physical attributes and their fitness.
As argued in the discussion of the previous requirement, the cladogram depicts a population of strategies. Therefore, diversity between the individuals that constitute the cladogram exists. Moreover, both the process of constructing it and the existing literature provide sufficient evidence of the satisfaction of this requirement.
The construction of the cladogram requires the collection of information from numerous organisations that implement strategies for the discovery of new drugs. Each organisation has a unique configuration and consequently a unique strategy. The configuration of each strategy depends on both genetic and environmental factors. The genetic factors include the direction and momentum generated by the individuals who form the organisation, the knowledge, and the experience. Environmental factors include legislation, diseases, competition, etc.

The collection of information for the construction of the cladogram provides evidence of variation on both the physical attributes and the fitness of drug discovery strategies. Individual organisations often pursue a unique strategy to achieve a competitive advantage through the discovery of new drugs. This is demonstrated by the expertise in certain therapeutic areas, which are part of the drug discovery strategy. Different organisations commit themselves to different therapeutic areas e.g. cardiovascular, oncology transplantation etc. Hence, it may be concluded that the physical attributes of their strategies vary.
The number and frequency of new drugs introduced to the market provides a measure of success for a drug discovery strategy. Each organisation and consequently each strategy have their own success rate. The difference between these success rates provides evidence for the variation and the fitness of strategies.

Heredity

Strategic reproduction takes place via the people that are employed and witness success or failure (Tsinopoulos, 2001). The main characteristics of successful strategies are being passed from organisation to organisation by education and imitation. Education provides a set amount of institutional knowledge, while imitation is associated with the copying of successful operating ways. As shown on the cladogram, new populations of strategies adopt the successful characteristics of their ancestors, shown as numbers on the body of the cladogram (Figure 2).
Those characteristics which are favoured by the environment have survived are present in the next generation of strategies. Those characteristics, however, which are not successful become obsolete or are unlearned (shown with a minus on the cladogram).

Natural Selection

The final requirement of evolution is that only the fittest strategies survive and hence are able to reproduce and transfer their successful characteristics to their descendants. As per the definition of the strategy for drug discovery, a surviving strategy will be one that *effectively materialises the corporation's goals and policies and manages to achieve competitive advantage*. Any drug discovery strategy that does not manage to achieve success will either be abandoned or drive the organisation that employs it to failure. In either of the two cases, unsuccessful strategies will cease to exist, while successful ones will survive and transfer their characteristics to their descendants. Using the cladogram of figure 2 the attributes of the successful strategies may be identified. For instance the strategy *Partnerships/Acquisition* has been successful over the last century (Cockburn et al., 2000, Henderson, 1994). The cladogram illustrates what are those characteristics that have made this strategy successful. Since these characteristics have not been removed for the development of later strategy (shown on the left hand-side of the cladogram) this strategy has been successful and thus selected in by the environment.

CONCLUSIONS

The aim of this paper was to explore the evolutionary characteristics of drug discovery strategies. These strategies are implemented by pharmaceutical organisations in an endeavour to discover new drugs and gain competitive advantage. Their change over time is the result of the development of new technology and the understanding of human

biology and chemistry. As argued in this paper the mechanism by which these strategies are changing is evolution. Using a cladistic classification, this paper has demonstrated how the four requirements of evolution, are met by strategies for drug discovery.
As revealed through this study the fittest strategies are those that allow for less emergence. Although *serendipity* is recognised as a fortunate accident, organisations cannot afford to 'gamble'. New technology and knowledge, to a great degree, aid organisations in targeting a disease and designing a drug to cure it. The classification presented in this paper shows how less emergent strategies are selected by the environment that surrounds them.

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